

XUEPENG FU

Postdoctoral Researcher, National Laboratory of the Rockies

isfuxp@gmail.com ♦ [Google Scholar](#) ♦ [Homepage](#) ♦ [GitHub](#)

ACADEMIC EMPLOYMENT

Postdoctoral Researcher in Mechanical Engineering

Apr. 2024–Present

National Laboratory of the Rockies, Golden, CO, USA

- Secured nearly \$800,000 in project funding as PI or co-PI, including a DOE Genesis project for which I lead a \$250,000 technical scope.
- Develop high-fidelity CFD/FSI and reduced-order models for wind-turbine aeroelasticity while leading validation of offshore wave-measurement systems.
- Led the development and wave-tank demonstration of an autonomous experimental system with six-degree-of-freedom motion, securing additional DOE funding.
- Lead cross-program collaboration and outreach across wind and water power, communicating results to DOE managers and industrial partners.

Research Assistant in Ocean Engineering

Jan. 2024–Mar. 2024

Shanghai Jiao Tong University, Shanghai, China

- Conducted experimental and computational investigations of vortex-induced vibration in flexible offshore structures and communicated research findings to project stakeholders.
- Mentored graduate and undergraduate students in experimental design, CFD modeling, data analysis, and research publication.

EDUCATION

Shanghai Jiao Tong University

Sept. 2018 - Dec. 2023

Ph.D. in Ocean Engineering (Advisor: Shixiao Fu, Outstanding thesis award)

Shanghai, China

Ocean University of China

Sept. 2014 - July 2018

B.Eng. in Ocean Engineering (Advisor: Weiping Huang, Outstanding thesis award)

Qingdao, China

RESEARCH INTERESTS

Fluid-structure interaction (especially vortex-induced vibration) in offshore oil & gas and renewable energy systems, investigated through experimental, numerical, and data-driven approaches.

PROJECTS EXPERIENCE

1. **AI-enabled Flow-Induced Vibration Modeling, Genesis Mission, Department of Energy, USA, 08/2026 - Present (Co-PI) [\$749,105, my-share: \$250,000]**
2. Research of Autonomous Wave Tank Experimental System, Open R&D Program, Department of Energy, USA, 09/2025 - Present (PI) [\$30,000, my-share: \$30,000]
3. U.S. Wave and Tidal Resource Characterization, Department of Energy, USA, 04/2026 - Present (Subtask Technical Lead)
4. Uncertainty Quantification of Wave Measurement, Department of Energy, USA, 04/2024 - Present (Technical Lead)
5. PacWave: Offshore Wave Energy Test Site, Department of Energy, USA, 04/2024 - Present (Subtask Technical Lead)
6. High-Fidelity Modeling, Exascale Computing Project, Department of Energy, USA, 04/2024 – 09/2025 (Subtask Technical Lead)

7. FIFERS - Flow Induced Force and its Effects on Risers, Joint Industry Program, DNV, Aker Solutions, Equinor, CNOOC, CPR Subsea, ABS, 10/2020 - 12/2023 (Technical Lead)

JOURNAL ARTICLES: FIRST/CORRESPONDING AUTHOR *: CORRESPONDING AUTHOR

1. **X. Fu***, F. Driscoll, R. Fao, C. Kenny, K. P. Griffin, M. Murphy & S. Lambert (2026). "Performance evaluation of an offshore wave measurement buoy in monochromatic waves." *Ocean Engineering*, 364, 127021.
2. **X. Fu**, S. Fu*, C. Liu, J. Shen & Z. Niu (2026). "Hydrodynamic properties of a flexible pipe in bidirectionally sheared flows undergoing vortex-induced vibration." *Journal of Fluid Mechanics*, 1034, A44.
3. **X. Fu***, G. Vijayakumar, B. Lee & M. Sprague (2026). "Numerical simulation of vortex-induced vibration response of a single IEA 10-MW wind turbine blade." *Journal of Fluids and Structures*, 143, 104537.
4. **X. Fu**, S. Fu*, Z. Niu, B. Zhao, J. Shen & P. Deng (2025). "A validated fluid-structure interaction simulation model for vortex-induced vibration of a flexible pipe in steady flow." *Marine Structures*, 104, 103895.
5. **X. Fu**, S. Fu*, Z. Han, Z. Niu, M. Zhang & B. Zhao (2025). "Numerical simulations of 2-DOF vortex-induced vibration of a circular cylinder in two and three dimensions: A comparison study." *Journal of Ocean Engineering and Science*, 10(4), 395–410.
6. **X. Fu**, S. Fu*, C. Liu, M. Zhang & Q. Hu (2024). "Data-driven approach for modeling Reynolds stress tensor with invariance preservation." *Computers & Fluids*, 274, 106215.
7. **X. Fu**, S. Fu*, M. Zhang, Y. Xu, H. Ren & T. Sun (2024). "Mean drag force of flexible riser under bidirectionally sheared flow." *Journal of Shanghai Jiao Tong University*, 58(11), 1637–1643.
8. **X. Fu**, S. Fu*, M. Zhang, H. Ren, B. Zhao & Y. Xu (2024). "Vortex-induced vibration of a flexible pipe under oscillatory sheared flow." *Physical Review Fluids*, 9(1), 014604.
9. **X. Fu**, S. Fu*, H. Ren, W. Xie, Y. Xu, M. Zhang, Z. Liu & S. Meng (2022). "Experimental investigation of vortex-induced vibration of a flexible pipe in bidirectionally sheared flow." *Journal of Fluids and Structures*, 114, 103722.
10. **X. Fu**, S. Fu*, M. Zhang, Z. Han, H. Ren, Y. Xu & B. Zhao (2022). "Frequency capture phenomenon in tandem cylinders with different diameters undergoing flow-induced vibration." *Physics of Fluids*, 34(8), 085120.
11. **X. Fu**, M. Zhang*, S. Fu, B. Zhao, H. Ren & Y. Xu (2022). "On the study of vortex-induced vibration of a straked pipe in bidirectionally sheared flow." *Ocean Engineering*, 266, 112945.
12. **X. Fu** & W. Huang* (2018). "Experimental study of out-of-plane motion of a steel catenary riser in rigid mode." *The Ocean Engineering*, 36(5), 114–120.

JOURNAL ARTICLES: CO-AUTHORED

*: CORRESPONDING AUTHOR

-
1. Z. Niu, S. Fu*, **X. Fu**, M. Zhang, J. Shen & P. Deng (2027). "Vortex-induced vibration suppression of a free-hanging riser using U-tail and short-tail fairings." *Marine Structures*, 111, 104167.
 2. P. Deng, J. Shen, R. Du, **X. Fu**, M. Zhang & S. Fu* (2026). "High-Reynolds-number effects on 2-DOF vortex-induced vibration: A hybrid experimental investigation." *Ocean Engineering*, 367, Part 4, 127901.
 3. Z. Niu, S. Fu*, **X. Fu**, M. Zhang, J. Shen & Z. Yao (2026). "Experimental evaluation of helical strakes for suppressing vortex-induced vibration of a free-hanging riser." *Applied Ocean Research*, 175, 105238.

4. J. Shen, S. Fu*, **X. Fu**, P. Deng, Z. Niu & T. Moan (2026). “Hydrodynamic coefficients of two stationary tandem cylinders at varying spacing ratios and Reynolds numbers.” *Physics of Fluids*, 38(8), 085114.
5. Z. Niu, S. Fu*, **X. Fu**, X. Liu, M. Zhang, J. Shen & P. Deng (2026). “Experimental investigation on vortex-induced vibration of a free-hanging flexible riser under different bottom clump weights.” *Ocean Engineering*, 353, 124778.
6. D. Wang, Q. Hu, X. Xing, H. Wu, J. Liu, **X. Fu** & C. Yang* (2026). “Thinking like clinicians: Ranking-based multi-instance learning for PPG-based hemodynamic fluctuation detection.” *Pattern Recognition*, 174, 112981.
7. J. Shen, S. Fu*, **X. Fu**, P. Deng, Z. Niu & T. Moan (2026). “A hybrid experimental investigation of structural parameter effects on 2-DOF vortex-induced vibrations of a circular cylinder.” *Journal of Ocean Engineering and Science*, 11(1), 191–203.
8. J. Shen, S. Fu*, **X. Fu**, T. Moan & S. Sævik (2025). “Flow-induced vibration of twin-pipe model with varying mass and damping: A study using virtual physical framework.” *Journal of Fluids and Structures*, 139, 104438.
9. J. Shen, P. Deng, S. Fu*, **X. Fu**, L. Song & Y. Xu (2025). “Enhanced vortex-induced vibration prediction of marine risers using excitation coefficient databases at high Reynolds numbers.” *Journal of Fluids and Structures*, 138, 104406.
10. B. Zhao, S. Fu*, M. Zhang, P. Deng & **X. Fu** (2025). “Frequency-domain prediction method for wake-induced vibrations of a downstream flexible riser.” *Journal of Vibration Engineering*, 38(7), 1405–1413.
11. J. Wang, S. Fu*, M. Zhang, Y. Xu, H. Ren, **X. Fu** & Z. Niu (2024). “Experimental study of vortex-induced vibration response characteristics of a free-hanging riser under multi-degree-of-freedom motion of a platform.” *Journal of Vibration Engineering*, 37(12), 2093–2102.
12. J. Wang, Y. Xu*, H. Ren, **X. Fu**, S. Fu & M. Zhang (2024). “Experimental study on vortex-induced vibration of a flexible free-hanging pipe in uniform flow.” *Journal of Ship Mechanics*, 28(9), 1405–1419.
13. B. Zhao, S. Fu*, P. Deng, M. Zhang, Y. Bai & **X. Fu** (2024). “Frequency-domain prediction method for vortex/wake-induced vibrations of double flexible risers in tandem arrangements.” *Ocean Engineering*, 297, 116942.
14. Z. Yang, Y. Xu*, J. Jing, **X. Fu**, B. Wang, H. Ren, M. Zhang & T. Sun (2023). “Investigation of physics-informed neural networks to reconstruct a flow field with high resolution.” *Journal of Marine Science and Engineering*, 11(11), 2045.
15. B. Zhao, M. Zhang*, S. Fu, **X. Fu**, T. Sun & B. Song (2023). “Experimental investigation on vortex/wake-induced force of double unequal-diameter flexible cylinders in tandem.” *Physics of Fluids*, 35(5), 055134.
16. B. Zhao, M. Zhang*, S. Fu, **X. Fu**, H. Ren & Y. Xu (2023). “Drag coefficients of double unequal-diameter flexible cylinders in tandem undergoing vortex/wake-induced vibrations.” *Ocean Engineering*, 270, 113642.
17. H. Ren, S. Fu*, B. Zhao, M. Zhang, Y. Xu, J. Shen, **X. Fu**, Z. Zhang & J. Huang (2022). “Hydrodynamic force model for flexible pipe based on energy competition and applications into flow-induced vibration prediction in uniform flow.” *Marine Structures*, 86, 103291.
18. H. Song, W. Huang*, **X. Fu**, H. Yan & S. Chang (2021). “Empirical model of the wake-induced lift force on a cylinder with low mass ratio.” *Marine Structures*, 80, 103081.
19. H. Ren, M. Zhang*, Y. Wang, Y. Xu, S. Fu, **X. Fu** & B. Zhao (2020). “Drag and added mass coefficients of a flexible pipe undergoing vortex-induced vibration in an oscillatory flow.” *Ocean Engineering*, 210,

107541.

20. B. Zhu, W. Huang*, X. Yao, J. Liu & **X. Fu** (2020). “Acceleration test of a steel catenary riser and its simulation.” *Journal of Harbin Engineering University*, 41(1), 60–66.
21. H. Song, W. Huang* & **X. Fu** (2020). “Vortex-induced vibration of flexible tandem cylinders based on model tests and numerical simulations.” *Journal of Vibration and Shock*, 39(6), 64–73.
22. X. Hu*, H. Xu, M. Zhang & **X. Fu** (2020). “Occurrence conditions and mitigation measures for special operating conditions of offshore platform mooring and positioning systems.” *Ship Engineering*, 42(S1), 457–459.
23. B. Zhu, W. Huang*, X. Yao, J. Liu & **X. Fu** (2019). “Numerical and experimental analysis of a steel lazy-wave riser based on Cable3D.” *Journal of Ship Mechanics*, 23(12), 1452–1462.
24. X. Yao, W. Huang*, S. Chang, J. Liu & **X. Fu** (2018). “Modal analysis and test verification for out-of-plane vortex-induced vibration of an SCR.” *Journal of Vibration and Shock*, 37(13), 78–84.

PROFESSIONAL SERVICE

1. **Early Career Editorial Board, *Journal of Fluids and Structures*** **Dec. 2025 - Present**
2. Reviewer for *J. Fluids Struct.*, *Mar. Struct.*, *Eng. Struct.*, *Ocean Eng.*, *Appl. Ocean Res.*, *Int. Commun. Heat Mass Transfer*, *Phys. Fluids*, *OMAE*, *ISOPE*, et al.

CONFERENCE PROCEEDINGS

1. **X. Fu**, R. Fao, C. Kenny & F. Driscoll (2026). Experimental investigation of frequency-domain wave energy estimation uncertainty via Large-Amplitude Motion Platform. In *Offshore Technology Conference*, D031S033R005.
2. Y. Bai, M. Zhang, S. Fu, Y. Chu, **X. Fu**, H. Ren & Y. Xu (2024). An experimental investigation on vortex-induced vibration of double risers in tandem under bidirectionally sheared flow. In *International Conference on Offshore Mechanics and Arctic Engineering*, 87844, V006T08A019.
3. **X. Fu**, S. Fu, H. Ren, M. Zhang, Y. Xu & B. Zhao (2021). Experimental investigation of vortex-induced vibration of a flexible pipe with helical strakes in oscillatory flow. In *Proceedings of the 2021 Academic Conference of the Testing Technology Group, Ship Mechanics Academic Committee, Chinese Society of Naval Architects and Marine Engineers*.
4. **X. Fu**, Y. Xu, M. Zhang, H. Ren, B. Zhao & S. Fu (2020). Numerical simulation of vortex-induced vibration of two tandem cylinders with different diameters under uniform flow. In *International Conference on Offshore Mechanics and Arctic Engineering*, 84409, V008T08A034a.
5. B. Zhao, M. Zhang, Y. Xu, H. Ren, **X. Fu**, S. Fu & C. Li (2020). Experimental study on interference response characteristics of triple flexible risers under uniform flow. In *International Ocean and Polar Engineering Conference*, ISOPE-I-20-2174.

SELECTED TALKS AND PRESENTATIONS

1. “Vortex-induced vibration of offshore slender structures”, *Newcastle University*, July, 2026, Virtual. (invited)
2. “Experimental investigation of frequency-domain wave energy estimation uncertainty via large-amplitude motion platform”, *Offshore Technology Conference 2026*, May, 2026, Houston, USA.
3. “Uncertainty estimation of wave buoy measurement via Large Amplitude Motion Platform”, *NREL Center for Energy Conversion and Storage Systems Tech Talk*, Oct., 2025, Denver, USA. (invited)

4. “Vortex-induced vibration of a flexible cylinder under bidirectionally sheared flow”, *76th Annual Meeting of the Division of Fluid Dynamics*, Nov., 2023, Washington DC, USA.
5. “Investigation on layout optimization of helical strakes on marine riser”, *42nd International Conference on Ocean, Offshore and Arctic Engineering*, June, 2023, Melbourne, Australia.
6. “Experimental investigation of vortex-induced vibration of a flexible pipe with helical strakes in oscillatory flow”, *Academic Conference of Chinese Society of Naval Architects in 2021*, Oct., 2021, Kunming, China. (Outstanding Presentation, Top 2%)
7. “Numerical simulation of vortex-induced vibration of two tandem cylinders with different diameters under uniform flow”, *39th International Conference on Ocean, Offshore and Arctic Engineering*, Aug., 2020, Virtual.

HONORS & AWARDS

Key Contributor Award at National Laboratory of the Rockies	2025
National Outstanding Doctoral Dissertation in Ocean Engineering (1 of 15 nationwide)	2025
Outstanding Doctoral Dissertation at Shanghai Jiao Tong University (1 of 15 university-wide)	2023
Weichai Power Scholarship	2022
Outstanding Undergraduate Graduates at Ocean University of China	2018
Outstanding Bachelor Thesis at Ocean University of China (top 5%)	2018
Science and Technology Innovation Scholarship at Ocean University of China	2015, 2016, 2017, 2018

TEACHING & MENTORING EXPERIENCE

Teaching Assistant, Hydroelasticity *Shanghai Jiao Tong University, 2021–2022*

Assisted Prof. Shixiao Fu with lecture preparation and delivery, connecting wave theory with structural mechanics. Guided laboratory sessions, held office hours, and helped prepare final examinations.

Student Research Mentor *Shanghai Jiao Tong University*

Co-mentored Ph.D., master’s, and undergraduate students under faculty supervision, providing guidance on research methods, computational modelling, and data analysis.

Ph.D. students: Jiawei Shen, Zhibo Niu, and Pengqian Deng.

Master’s student: Jing Wang.

Undergraduate students: Yifan Wang and Junhui Wu.

PATENTS

1. H. Ren, S. Fu, **X. Fu**, J. Shen, M. Zhang, Y. Xu, T. Hu, B. Song, T. Sun (2025). Experimental setup for vortex-induced vibration of a riser in a bidirectional shear flow with flexible transmission. CN116625617A. (Chinese Patent)
2. H. Ren, Y. Xu, S. Fu, M. Zhang, **X. Fu**, J. Shen, B. Song, T. Sun (2025). Experimental setup for vortex-induced vibration of a riser in a bidirectional shear flow with flexible transmission. CN116519240A. (Chinese Patent)
3. **X. Fu**, W. Huang, C. Guo, Q. Wu, T. Yang, A. Liu, H. Song (2018). A method for determining the vibration response frequency of tandem cylinders in the wake. CN108168684A. (Chinese Patent)
4. **X. Fu**, M. Zhang, Q. Wu, Z. Jiang, C. Guo, A. Chang, W. Ma (2017). A movable multifunctional modular floating island group. CN106494587B. (Chinese Patent)